

6.3.2 Spectroscopy

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
NMR Spectroscopy	
(a) analysis of a carbon-13 NMR spectrum of an organic molecule to make predictions about: (i) the number of carbon environments in the molecule (ii) the different types of carbon environment present, from chemical shift values (iii) possible structures for the molecule	<i>M3.1</i> All carbon-13 NMR spectra that are assessed will be proton decoupled. In examinations, NMR chemical shift values will be provided on the <i>Data Sheet</i> . Restricted to functional groups studied in the A level specification. HSW3,5 Interpretation of spectra to analyse organic compounds.
(b) analysis of a high resolution proton NMR spectrum of an organic molecule to make predictions about: (i) the number of proton environments in the molecule (ii) the different types of proton environment present, from chemical shift values (iii) the relative numbers of each type of proton present from relative peak areas, using integration traces or ratio numbers, when required (iv) the number of non-equivalent protons adjacent to a given proton from the spin-spin splitting pattern, using the $n + 1$ rule (v) possible structures for the molecule	<i>M3.1</i> In examinations, NMR chemical shift values will be provided on the <i>Data Sheet</i> . Restricted to functional groups studied in the A level specification. Learners will be expected to identify aromatic protons from chemical shift values but will not be expected to analyse their splitting patterns. HSW3,5 Interpretation of spectra to analyse organic compounds.
(c) prediction of a carbon-13 or proton NMR spectrum for a given molecule	<i>M3.1</i>

- (d) (i) the use of tetramethylsilane, TMS, as the standard for chemical shift measurements
- (ii) the need for deuterated solvents, e.g. CDCl_3 , when running an NMR spectrum
- (iii) the identification of O–H and N–H protons by proton exchange using D_2O

Combined techniques

- (e) deduction of the structures of organic compounds from different analytical data including:

- (i) elemental analysis (**see also 2.1.3 c**)
- (ii) mass spectra (**see also 4.2.4 f–g**)
- (iii) IR spectra (**see also 4.2.4 d–e**)
- (iv) NMR spectra.

M3.1

Spectral reference data will be provided on the *Data Sheet*.

Restricted to functional groups studied in the A level specification.

Learners will **not** be expected to interpret mass spectra of organic halogen compounds.

HSW3,5,6 Interpretation of a variety of different evidence to analyse organic compounds.